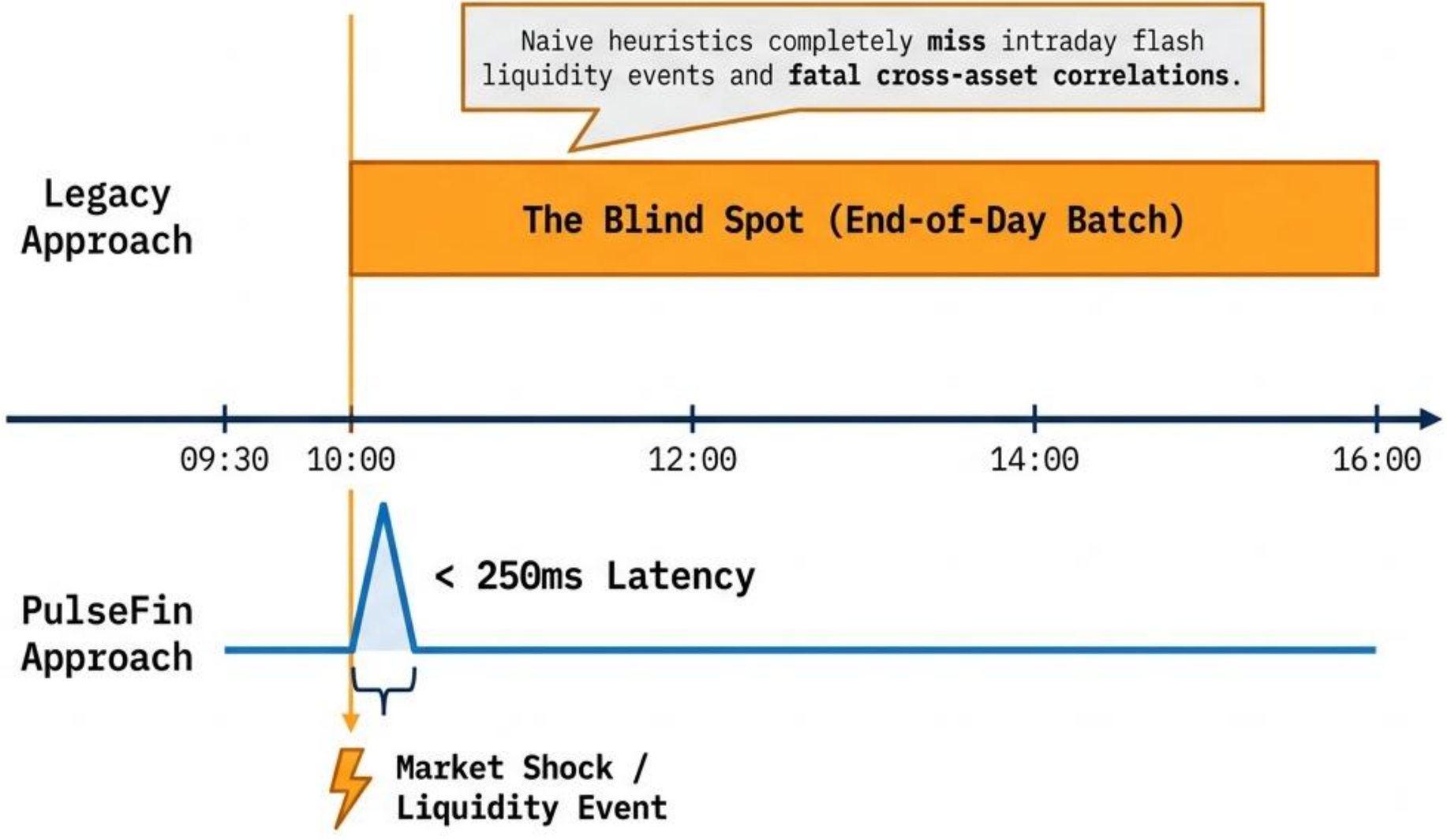
PulseFin

Bridging Unstructured Financial NLP with Peer-Reviewed Mathematical Portfolio Optimization

Target Integration: S&P Global & CRISIL Campus Hackathon 2026	
Lead Architect:	Manish Pawar
Origin Node:	BITS Pilani

AUTONOMOUS STRESS TESTING | LEDOIT-WOLF COVARIANCE SHRINKAGE
BLACK-LITTERMAN REBALANCING | ROCKAFELLAR-URYASEV CVaR HEDGING

Bridging the Intraday Risk Blind Spot



The Autonomous Pipeline

Ingests unstructured news in real time.

↓

Extracts sentiment & classifies systemic risk via **Transformers** (**finbert** + **mnli**).

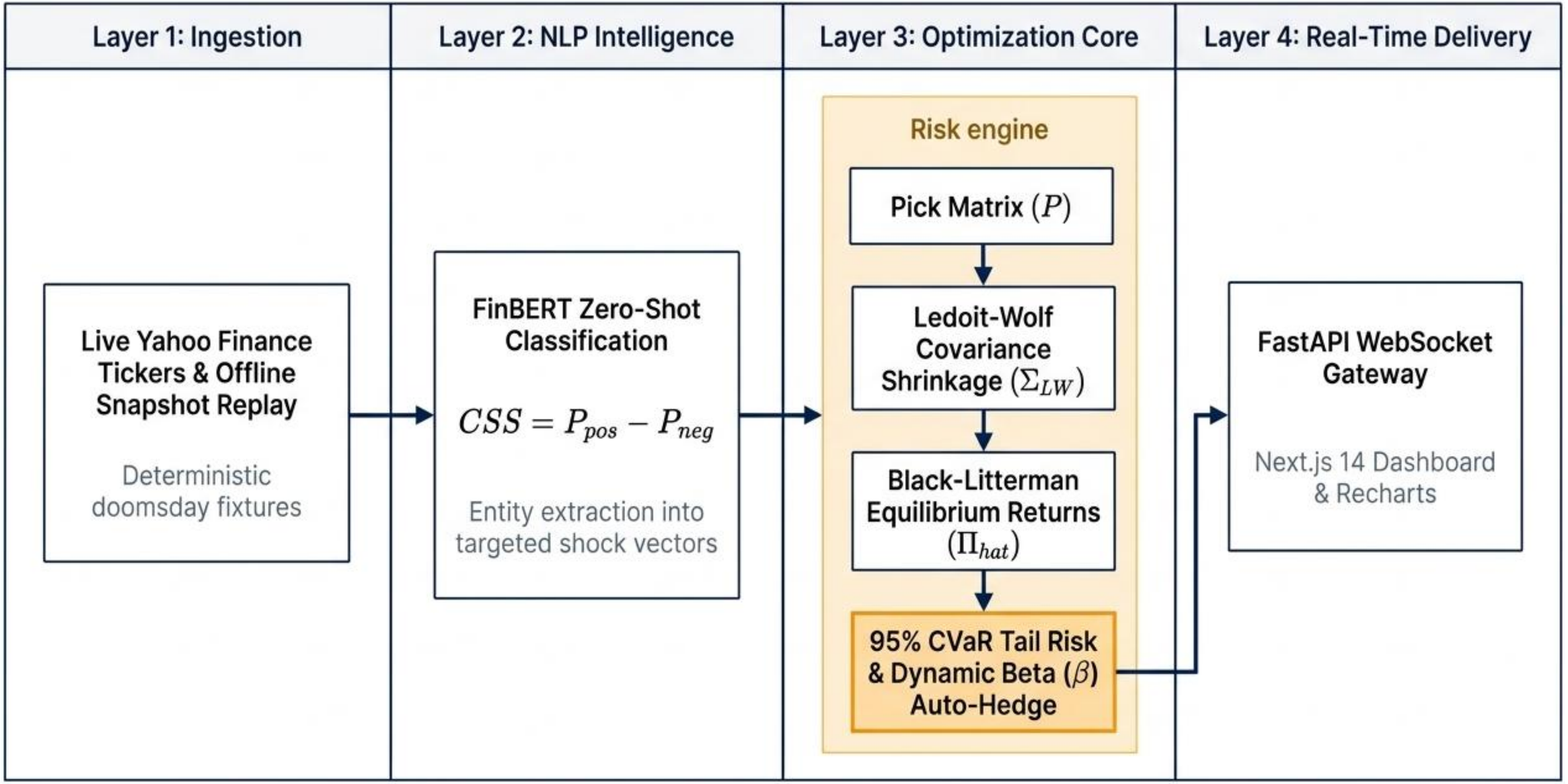
↓

Dynamically constructs targeted **Pick Matrices** (**P**).

↓

Streams actionable **Black-Litterman** rebalancing & **CVaR** auto-hedges over **WebSockets**.

End-to-End Mathematical Pipeline



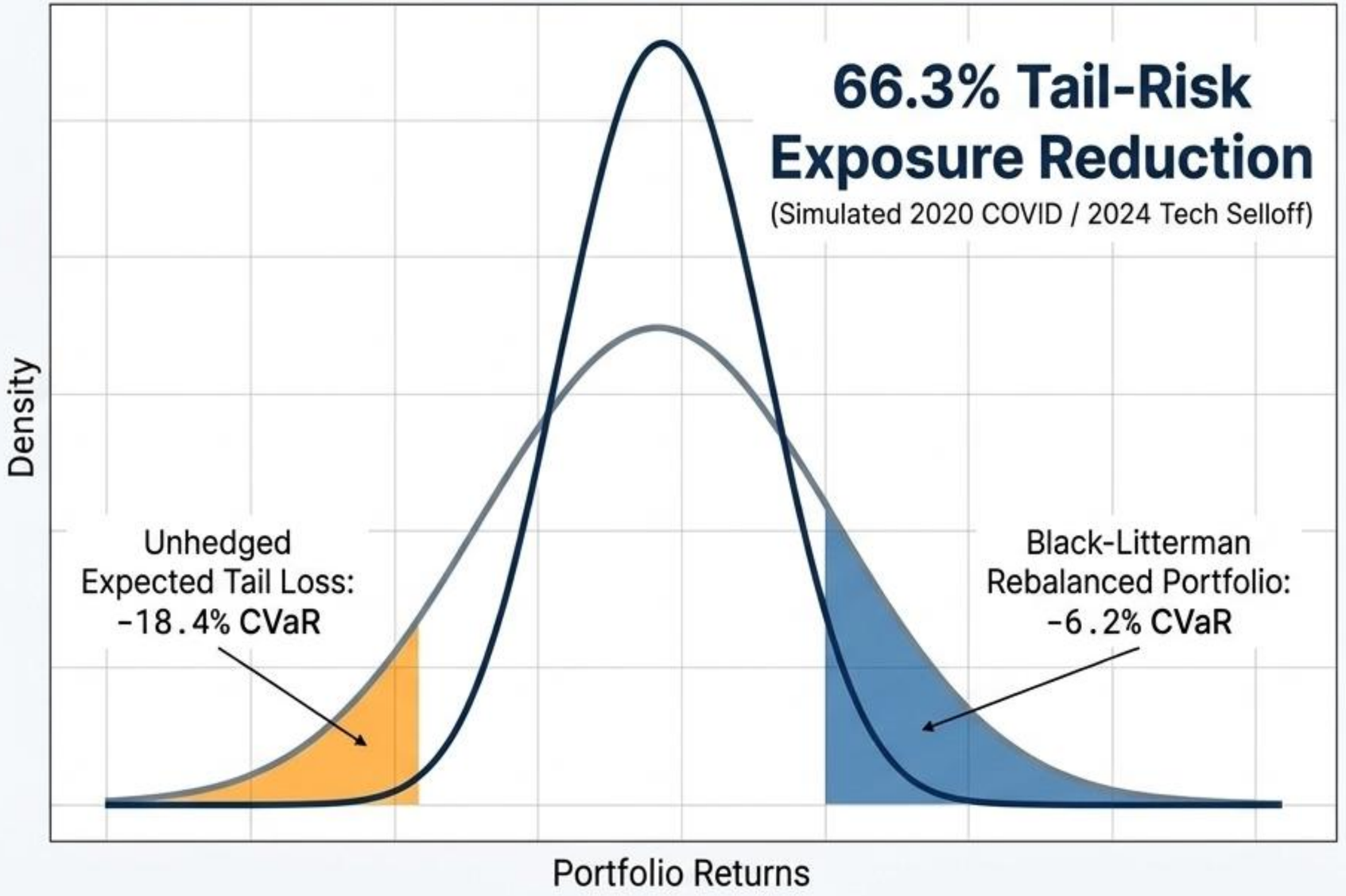
The Quantitative Edge: Architectural Moat

Architecture Component	Legacy Standard	PulseFin Engine (The Quant Edge)
NLP Intelligence	VADER Lexicons / Naive Bayes Secondary saxing failure	FinBERT ↑ +34% F1-score improvement. Accurately scores "Company cuts debt" as positive (+0.85), bypassing lexicon failures.
Covariance Math	Markowitz Sample Covariance	Ledoit-Wolf Shrinkage Optimally shrinks sample covariance toward identity matrix, preventing noise amplification and guaranteeing numerical stability.
Portfolio Allocation	Naive Rule Allocation	Dynamic Black-Litterman Blends market equilibrium (Π) with NLP views (Q, P), weighted precisely by confidence uncertainty (Ω).

Explainable AI (XAI)

All calculations produce human-auditable risk rationale traces for investment committees and risk officers.

Quantitative Benchmarks: Latency & Tail-Risk Mitigation



Pipeline Latency

< 250ms

News ingestion to full CVaR & rebalance calculation.

Delivery Latency

< 15ms

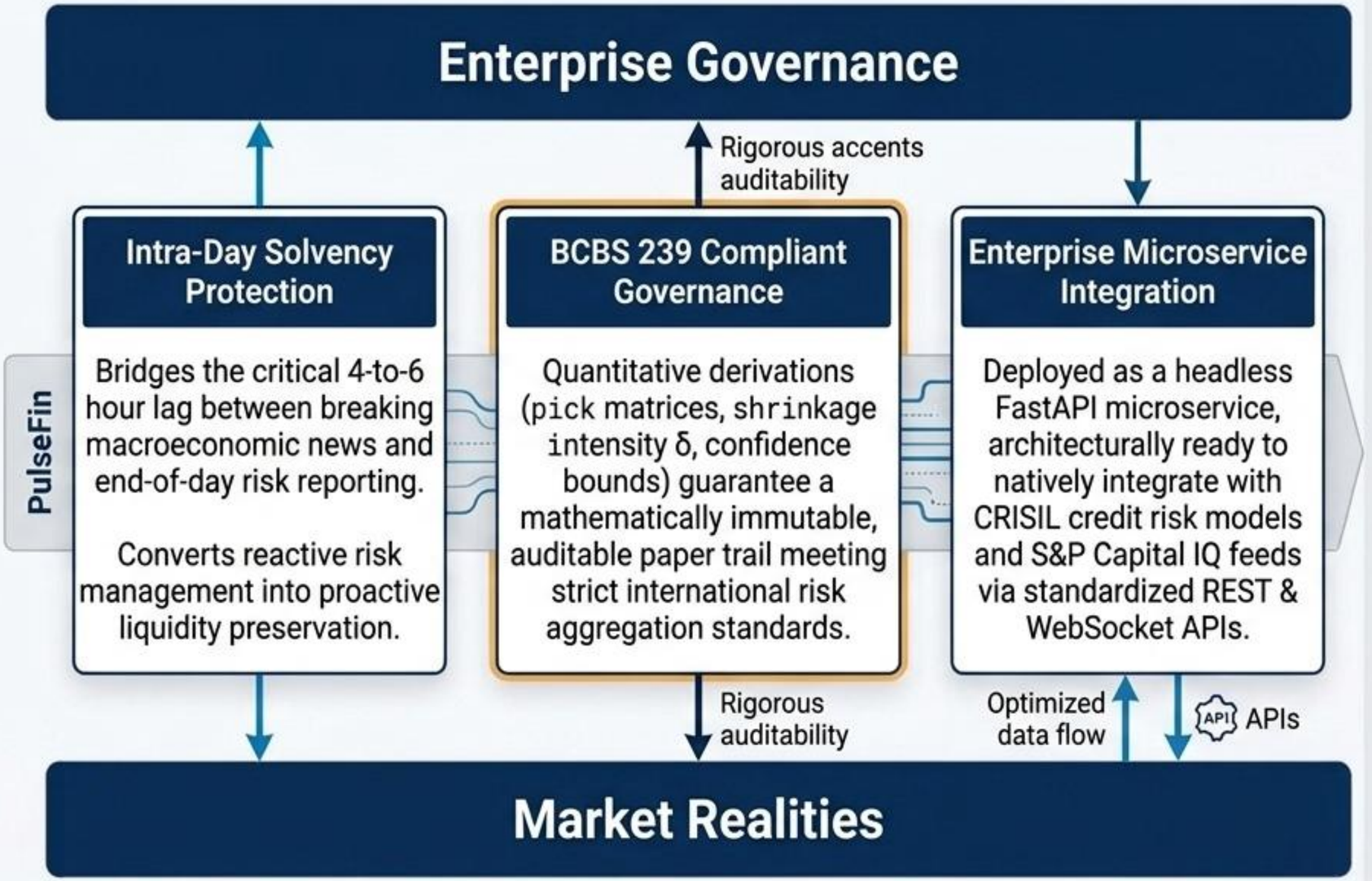
FastAPI WebSocket broadcast latency.

Auto-Hedging Execution

$\beta = 1.34$ to $\beta = 0.00$

Instant SPY put option hedge neutralizing portfolio beta to Delta-Neutral.

Synthesizing Business Value for S&P & CRISIL



Infrastructure Roadmap & Institutional Extensions

